

Exact Total Domination Numbers of Cylindrical Grids of Widths Five, Six, and Seven

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Abstract

We determine the total domination number of the cylindrical grids $P_m \square C_n$ for every circumference $n \geq 3$ when $m \in \{5, 6, 7\}$. The proof represents a total dominating set by a closed walk in a finite column automaton whose states record selected rows and the rows still requiring domination from the next column. The graph parameter is thereby the minimum diagonal entry of a min-plus matrix power. For widths five, six, and seven we verify exact entrywise matrix identities with periods four, fourteen, and four, respectively. Exact finite certificates supplied with the paper establish these identities and the finite prefix values needed to complete the computer-assisted part of the proof. The resulting formulas include the exceptional circumferences $n = 12$ for width six and $n = 7, 14$ for width seven.

Keywords: total domination; cylindrical grid; Cartesian product; finite-state method; min-plus algebra; computer-assisted proof.

1 Introduction

Let G be a graph without isolated vertices. A set $D \subseteq V(G)$ is *totally dominating* if every vertex of G , including every vertex of D , has a neighbor in D . The minimum cardinality of such a set is the total domination number $\gamma_t(G)$. In this paper we study the cylindrical grids $P_m \square C_n$, the Cartesian product of a path and a cycle.

Total domination in grid-like products has a substantial literature. Gravier studied total domination in products of paths and cycles and related tiling formulations [1]. Exact cylindrical results for smaller fixed path widths were obtained by Eakawinrujee for widths two, three, and four [3]; the author's dissertation gives additional provenance for that cylinder program [4]. Rectangular grids, ordinary domination, and several variants provide nearby but distinct problems; for example, the fixed-width results of Crevals and Ostergard concern $P_m \square P_n$, not cylinders [2], while Nandi, Parui, and Adhikari study ordinary domination [6].

The results below concern widths not covered by the fixed-width cylindrical formulas cited above. Hu, Sohn, and Chen also report exact total-domination values for earlier bundle cases, including the small-width overlap and $P_5 \square C_n$ [5]. We therefore make the narrow claim of complete classifications for widths five, six, and seven using one certified finite-state proof package, without claiming priority over those earlier cases. The cyclic boundary condition couples the first and last columns, and a selected vertex cannot dominate itself. Our approach handles both issues with a finite-state representation; the semantic reduction is proved in the paper and the finite identities are supplied as exact computer-assisted certificates.

The methodological distinction from recent nearby work is also important. Martínez, Castaño-Fernández, and Puertas use tropical matrix products for 2-domination of cylinders [7]; that invariant requires two selected neighbors, whereas ordinary total domination requires one open neighbor for every vertex, including selected vertices. The shared min-plus language is methodological context, not an equivalence of parameters.

Our main results are the following.

Theorem 1.1 (Width five). For every $n \geq 3$,

$$\gamma_t(P_5 \square C_n) = \begin{cases} \lceil 3n/2 \rceil + 1, & n \equiv 2 \pmod{4}, \\ \lceil 3n/2 \rceil, & \text{otherwise.} \end{cases}$$

Theorem 1.2 (Width six). Let ε_r , for $r \in \{0, \dots, 13\}$, be given by

$$\varepsilon_r = \begin{cases} 0, & r \in \{0, 3, 5, 7, 9, 10\}, \\ 1, & r \in \{1, 4, 6, 11, 13\}, \\ 2, & r \in \{2, 8\}, \\ 3, & r = 12. \end{cases}$$

Then

$$\gamma_t(P_6 \square C_n) = \left\lceil \frac{12n}{7} \right\rceil + \varepsilon_{n \bmod 14}$$

for every $n \geq 3$, except that $\gamma_t(P_6 \square C_{12}) = 22$.

Theorem 1.3 (Width seven). For every $n \geq 3$,

$$\gamma_t(P_7 \square C_n) = 2n,$$

except $\gamma_t(P_7 \square C_7) = 15$ and $\gamma_t(P_7 \square C_{14}) = 30$.

The proof has two layers. First, we prove the graph-theoretic equivalence between total dominating sets and weighted closed walks. Second, a finite certificate establishes the matrix identities and prefix values. The thresholded recurrence lemma then shows exactly how those finite facts imply each all-circumference formula.

For orientation, the following small-width values are contextual overlap with Hu–Sohn–Chen and are not claimed as new here:

m	$\gamma_t(P_m \square C_3)$	$\gamma_t(P_m \square C_4)$
5	5	6
6	6	8
7	6	8

2 Preliminaries

We use $[m] = \{0, 1, \dots, m-1\}$ for the rows of P_m , and index the cycle columns by $\mathbb{Z}/n\mathbb{Z}$. For $S \subseteq [m]$, define its open path neighborhood by

$$N_P(S) = \{i \in [m] : i-1 \in S \text{ or } i+1 \in S\},$$

where indices outside $[m]$ are omitted. Thus a row selected in a column does not automatically dominate the vertex in that same row and column.

For a set $D \subseteq V(P_m \square C_n)$, write

$$S_j = \{i \in [m] : (i, j) \in D\}.$$

The horizontal neighbors of (i, j) are $(i, j-1)$ and $(i, j+1)$, and the vertical neighbors are the vertices in rows adjacent to i in the same column.

3 The finite-state model

For a cyclic sequence of selected row sets (S_j) , define the pending set

$$R_j = [m] \setminus (S_{j-1} \cup N_P(S_j)).$$

The rows in R_j are precisely those for which the left horizontal and vertical domination alternatives are absent in column j . Define the fixed admissible state set

$$Q_m = \{(S, R) : S \subseteq [m], R \subseteq [m] \setminus N_P(S), \text{ and } R = [m] \setminus (U \cup N_P(S)) \text{ for some } U \subseteq [m]\}.$$

Thus the predecessor mask is quantified in the definition but is not part of the state. Given a state $q = (S, R) \in Q_m$ and an input mask $T \subseteq [m]$, the transition is legal precisely when

$$R \subseteq T,$$

and its head pending set is forced to be

$$R' = [m] \setminus (S \cup N_P(T)).$$

The edge is $(S, R) \rightarrow (T, R')$, where the displayed formula defines R' , provided $R \subseteq T$ and $(T, R') \in Q_m$; its weight is $|T|$. Every edge head is therefore a member of the fixed finite set Q_m .

Theorem 3.1 (Finite-state representation). For $m \in \{5, 6, 7\}$ and $n \geq 3$, labeled total dominating sets of $P_m \square C_n$ are in weight-preserving bijection with labeled length- n closed walks in the state graph above. The weight of a walk is the sum of its transition weights.

Proof. We first prove soundness. Let a cyclic legal walk be given and define $D = \{(i, j) : i \in S_j\}$. Consider a vertex (i, j) . If $i \notin R_j$, then by the definition of R_j , either $i \in S_{j-1}$ or $i \in N_P(S_j)$. Hence (i, j) has a selected left or vertical open neighbor. If $i \in R_j$, transition legality gives $i \in S_{j+1}$, so the right horizontal neighbor is selected. These cases include selected vertices themselves and therefore establish total domination.

Conversely, let D be a total dominating set and define S_j from its selected vertices and R_j by the displayed formula. If $i \in R_j$, then the left and every vertical selected-neighbor alternative is absent. Total domination forces the right neighbor $(i, j+1)$ to be selected, so $R_j \subseteq S_{j+1}$. The next pending formula follows from its definition. The same argument applies to the wraparound transition.

The two constructions are inverse because the pending masks are forced by the selected masks. Finally,

$$\sum_j |S_{j+1}| = \sum_j |S_j| = |D|,$$

so weights agree. For $n = 3$, the left and right horizontal neighbors are distinct because $3 \nmid 2$, so the cyclic encoding has exactly the simple graph neighborhood semantics. $\square$

4 Min-plus path semantics and the recurrence certificate

Let M_m be the weighted adjacency matrix of this graph over $\mathbb{Z}_{\geq 0} \cup \{+\infty\}$, with $+\infty$ for a missing edge. Min-plus addition and multiplication are

$$a \oplus b = \min(a, b), \quad a \otimes b = a + b,$$

with $+\infty$ absorbing for $\otimes$. Matrix powers use min-plus multiplication. The min-plus identity matrix has zero diagonal and $+\infty$ off the diagonal. For $c \in \mathbb{Z}_{\geq 0}$, scalar matrix shift means $(c \otimes A)_{uv} = c + A_{uv}$ for finite A_{uv} , and $(c \otimes A)_{uv} = +\infty$ otherwise.

Lemma 4.1 (Min-plus path semantics). For every $k \geq 0$, $(M_m^k)_{uv}$ is the minimum weight of a length- k walk from u to v , with the minimum of an empty set interpreted as $+\infty$.

Proof. The case $k = 0$ is the min-plus identity matrix. For the induction step, partition a length- $k + 1$ walk by its penultimate state x . The prefix has minimum cost $(M_m^k)_{ux}$, the final edge has cost $(M_m)_{xv}$, and minimizing over the finite state set gives the min-plus product. Missing prefixes or edges contribute $+\infty$ and therefore do not create a walk. $\square$

Define

$$g_m(n) = \min_{q \in Q_m} (M_m^n)_{qq}.$$

Theorem 3.1 and Lemma 4.1 give

$$\gamma_t(P_m \square C_n) = g_m(n), \quad n \geq 3.$$

Lemma 4.2 (Thresholded min-plus recurrence). Suppose an entrywise identity

$$M^{N+p} = c \otimes M^N$$

holds for a finite integer c . Define $h(t) = \min_q (M^t)_{qq}$. Then for every $k \geq 0$,

$$M^{N+p+k} = c \otimes M^{N+k},$$

and hence $h(n + p) = h(n) + c$ for every finite-value base $n \geq N$.

Proof. Min-plus matrix multiplication is associative: both parenthesizations minimize the same finite collection of triple products. A finite scalar shift commutes with a subsequent min-plus product, including entries equal to $+\infty$. Therefore, for $k \geq 0$,

$$M^{N+p+k} = M^{N+p} \otimes M^k = (c \otimes M^N) \otimes M^k = c \otimes (M^N \otimes M^k) = c \otimes M^{N+k}.$$

Taking the minimum over diagonal entries gives the same statement for g . When the relevant diagonal minimum is finite, the scalar operation is ordinary addition. The conclusion is explicitly restricted to $n = N + k \geq N$; no value below the threshold is propagated by this lemma. $\square$

5 Finite certificates and exact formulas

The finite certificate proposition below is the computational input to the three theorem proofs. Its verifier reconstructs the state and transition semantics, parses the persisted matrices, recomputes both powers, and checks the entrywise identities with explicit $+\infty$ semantics. The proposition is computer-assisted: its trusted inputs are the definitions, the certificate files, and the verifier specified in the reproducibility record.

Proposition 5.1 (Finite certificate and prefix data). For widths five, six, and seven, the state and transition graphs have the counts in Table 1; the displayed matrix identities hold; and the diagonal minima $g_m(n)$ agree with the exact finite-prefix tables in the project artifact on $3 \leq n \leq 19$, $3 \leq n \leq 34$, and $3 \leq n \leq 31$, respectively.

The certified identities are

$$M_5^{20} = 6 \otimes M_5^{16}, \quad M_6^{35} = 24 \otimes M_6^{21}, \quad M_7^{32} = 8 \otimes M_7^{28}.$$

Consequently the recurrences begin at $n = 16$, $n = 21$, and $n = 28$, respectively. The exact finite-prefix replay covers $3 \leq n \leq 19$ for width five, $3 \leq n \leq 34$ for width six, and $3 \leq n \leq 31$ for width seven. Since each interval ends at $N + p - 1$, every later index reduces to a checked base in $[N, N + p - 1]$.

The decisive finite values are displayed here. The complete machine-readable tables, including all rows from 3 through the stated prefix endpoint, are part of the reproducibility record.

Table 1: Finite state graphs and entrywise recurrence certificates.

width	states	transitions	N	p	c
5	169	2,419	16	4	6
6	441	11,025	21	14	24
7	1,156	50,303	28	4	8

Table 2: Width-five prefix values.

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$g_5(n)$	5	6	8	10	11	12	14	16	17	18	20	22	23	24	26	28	29

5.1 Width five

The identity $M_5^{20} = 6 \otimes M_5^{16}$ gives $g_5(n+4) = g_5(n) + 6$ for $n \geq 16$. *Proof of Theorem 1.1.* The finite proposition gives every value $3 \leq n \leq 19$; the base values $16 \leq n \leq 19$ are shown in Table 2, and the complete prefix comparison is in the machine-readable table. Every $n \geq 20$ reduces by four to one of these bases, and the residue $n \equiv 2 \pmod{4}$ contributes the additional unit.

5.2 Width six

The identity $M_6^{35} = 24 \otimes M_6^{21}$ gives $g_6(n+14) = g_6(n) + 24$ only for $n \geq 21$. *Proof of Theorem 1.2.* The value at $n = 12$ is a finite-prefix exception and is not propagated to $n = 26$. The finite proposition gives every value $3 \leq n \leq 34$, including all rows $3 \leq n < 21$ and every base value $21 \leq n \leq 34$. For every $n \geq 35$, repeated subtraction of fourteen reaches one of the checked values $21, \dots, 34$, which proves Theorem 1.2; the recurrence is never applied below its threshold.

5.3 Width seven

The identity $M_7^{32} = 8 \otimes M_7^{28}$ gives $g_7(n+4) = g_7(n) + 8$ only for $n \geq 28$. *Proof of Theorem 1.3.* The finite proposition explicitly covers every row $3 \leq n \leq 31$, including the ordinary prefix rows and the exceptions $g_7(7) = 15$ and $g_7(14) = 30$. Every $n \geq 32$ reduces to a base in $28, \dots, 31$, proving the theorem. The exceptions remain only in the finite prefix and are not propagated by the recurrence.

6 Verification and reproducibility

The repository includes a reference implementation and a separately written certificate verifier. The verifier rebuilds states, transitions, weights, and matrix powers from the definitions. Detailed test counts, mutation results, certificate manifests, and hash bindings are recorded separately from the mathematical proof. The three certificate commands are:

```
python verify/verify_certificates.py 5 certificates/width5
python verify/verify_certificates.py 6 certificates/width6
python verify/verify_certificates.py 7 certificates/width7
```

Each command reports zero identity mismatches. The repository’s SHA256SUMS file binds the finite artifacts, and paper/REPRODUCIBILITY.md records the expected dimensions, counts, parameters, and verification outputs.

The computations establish the finite proposition; the preceding semantic lemmas and theorem arguments establish why that proposition implies the infinite statements.

Table 3: Width-six base and exceptional values.

n	12	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$g_6(n)$	22	36	40	40	42	44	48	48	48	51	54	54	56	57	60

Table 4: Width-seven exceptional and tail-base values.

n	7	14	28	29	30	31
$g_7(n)$	15	30	56	58	60	62

7 Discussion

The three widths have different eventual periods: four for widths five and seven, and fourteen for width six. This difference is visible in the finite state graphs and in the residue correction for width six. The method scales naturally as a fixed-width transfer construction, but the finite state space grows rapidly with the width. No claim about width eight is made here.

A cyclic domination problem can be expressed as a closed-walk problem without sacrificing the open-neighborhood condition. Once the finite weighted graph is fixed, an entrywise min-plus identity controls every subsequent walk length at and above its threshold; the finite prefix accounts for the transients and exceptional circumferences.

8 Conclusion

We determined the exact total domination numbers of $P_5 \square C_n$, $P_6 \square C_n$, and $P_7 \square C_n$ for every $n \geq 3$. The proof combines a weight-preserving finite-state bijection, min-plus path semantics, three exact entrywise matrix identities, and finite-prefix verification. The computer-assisted portion is specified by the finite proposition and the reproducibility record, while the graph-theoretic reduction and recurrence argument are given explicitly above.

References

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